Lab 5 - Static Code Analysis: Issues Table

File: hospital_patient_system.py. Line numbers are from the original file.
Pylint: 5.19/10 before, 10.00/10 after. Bandit: 5 issues before, 0 after. Flake8: 11 warnings before, 0 after.

Issue	Type	Line(s)	Description	Fix Approach
eval() used (Bandit B307, Medium)	Security	40	eval runs whatever string is passed in, so anyone could run their own code	Removed eval. The function now takes a number (rate) and returns base_bill * rate
os.system with filename (Bandit B605, High)	Security	56	filename is added into a shell command, so shell injection is possible	Removed it and used print() instead
MD5 hash (Bandit B324, High)	Security	45	md5 is a weak hash	Changed to sha256
random for token (Bandit B311, Low)	Security	49	random module is predictable	Used secrets.randbelow(9000) + 1000
Bare except with pass (Bandit B110, Flake8 E722, Pylint W0702)	Bug	22-25	catches every error and hides it, and an unknown patient is silently ignored	Check if p_id exists first and return True or False
Hardcoded access key (not flagged by tools)	Security	68, 77	secret is written in the code and also printed	Read it from the NURSE_ACCESS_KEY environment variable and stopped printing it
Mutable default argument (Pylint W0102)	Bug	13	intake_log=[] is shared across calls	Default is None, uses a module level INTAKE_LOG
Float output 1344.0000000000002 (not flagged by tools)	Bug	72	cost is not rounded	Used round(..., 2)
File opened without with (Pylint R1732, W1514)	Bug	53-55	file may stay open if write fails, no encoding given	Used with open(..., encoding="utf-8")
main() runs on import (not flagged by tools)	Design	80	the demo runs as soon as the file is imported	Added if __name__ == "__main__":
Unused import and variables (Flake8 F401, F841, Pylint W0611, W0612)	Quality	4, 29, 39, 61	math, bed_charge and p are never used	Removed them (base_bill is now used)
Comparison to None with == (Flake8 E711, Pylint C0121)	Style	30	should use is None	Changed to is None
Unneeded if/else returning True/False (Pylint R1703, R1705)	Quality	32-35	longer than it needs to be	Replaced with one return statement
Parameter hides global variable (Pylint W0621)	Quality	59	patient_records parameter has the same name as the global	Renamed the parameter to records
Function name admitPatient (Pylint C0103)	Style	13	not snake_case	Renamed to admit_patient
% string formatting (Pylint C0209)	Style	17	can be an f-string	Changed to an f-string
Missing docstrings (Pylint C0114, C0116)	Style	whole file	module and all 9 functions had none	Added short docstrings
Lines too long (Flake8 E501)	Style	7-9, 16, 72	more than 79 characters	Split the lines
No newline at end of file (Flake8 W292, Pylint C0304)	Style	80	last line has no newline	Added one